Certified feasible-capacity screening for interconnection queues

Method, results on IEEE test systems, and a proposed offline pilot on one historical queue cycle.

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All results: simulation on public IEEE cases (case30, case118)

A millisecond screen that says which of its own answers can be trusted.

SPEED

1.40 ms

per request, against a median 390 AC power flows for a verified answer (case30).

CERTIFICATE

91.0%

coverage on 311 held-out requests at a 90% target.

SAFETY

47.9 → 9.0%

overestimates of capacity, raw learned estimate vs certified bound.

WHERE IT BREAKS

66.3%

coverage when demand shifts outside training. Shown on slide 09.

Anything the certificate can't clear goes to the normal AC study. The screen changes the order of studies, never the decision.

Study cost is (rounds) × (outages) power flows per request, times the queue.

- A request scales an injection pattern **u** by **α**. Its capacity is the largest α at which the intact network and every studied single-branch outage stay within thermal and voltage limits.
- Finding it is a bisection over α. Every step runs **one AC power flow per outage**.
- On case30 with 38 studied outages, one verified label costs a median **390 AC power flows** (0.89 s).
- The application side of the queue (intake, completeness) is being addressed elsewhere, including PJM's work with Google and Tapestry. This targets the study step.

OUTPUTS PER REQUEST

1. Linear N-1 capacity and the binding (outage, branch) pair
2. Learned estimate of the AC capacity
3. Certified lower bound on AC capacity at a chosen coverage level
4. A defer flag: requests the certificate can't clear go to the full AC study

Queue background: LBNL, Queued Up (2024)

Three components. The screen orders the studies. It never replaces them.

1 · LINEAR SECURITY LAYER

Exact DC N-1 capacity

Injection shift factors for the intact network and each outage (rank-one updates from outage distribution factors, no re-inversion). Closed-form capacity and its binding constraint.

2 · POOLED AC LABELER

Verified AC labels

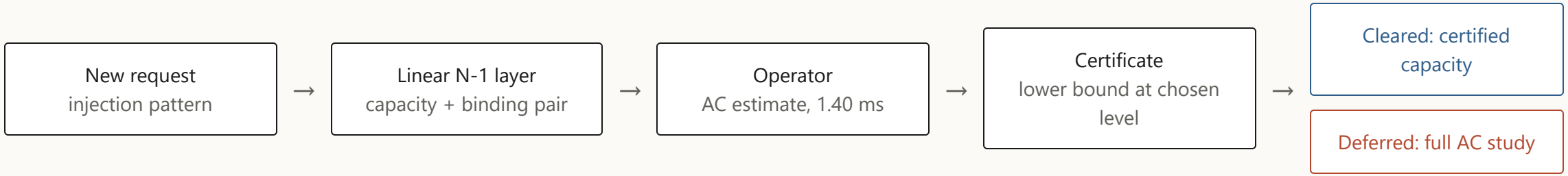
Batched Newton solves across outages and requests. Each request keeps its own bisection bracket. Non-convergence counts as insecure, so labels are lower bounds.

3 · OPERATOR + CERTIFICATE

Fast AC estimate

A graph DeepONet predicts the AC/DC derate from the linear layer's sensitivities. Split-conformal calibration gives a certified lower bound and a defer rule.

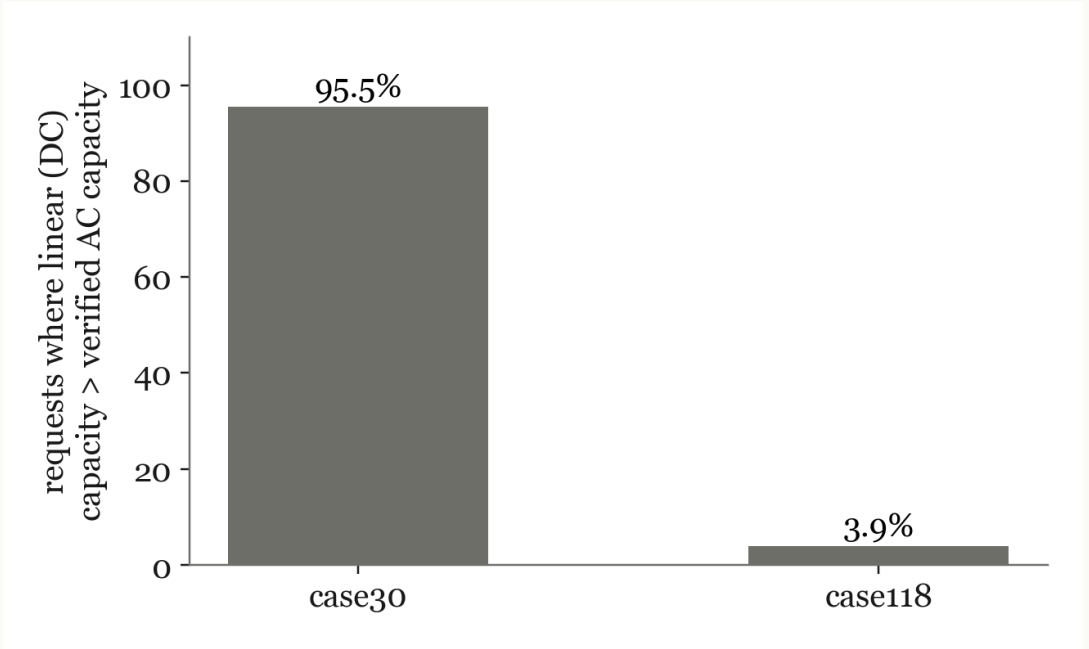
Training labels come from component 2. At run time only components 1 and 3 execute, which takes milliseconds per request.



$$a^{(c)} = H^{(c)}\, p_0 \qquad b^{(c)} = H^{(c)}\, u$$

$$\alpha^* = \min \text{ over } c, \ell \text{ of } \min\left((\bar{f}_\ell - a_\ell^{(c)}) / b_\ell^{(c)} ,\ (\bar{f}_\ell + a_\ell^{(c)}) / (-b_\ell^{(c)}) \right)$$

- The minimizer is the binding (outage, branch) pair.
- Shift factors match pandapower makePTDF to **2.7e-15** (case14, 30, 57). The closed form matches a dense scan to **7.4e-07**.
- DC branch flows track AC to 4.6% median error (16.5% at p95). But DC has no voltage limits and no losses.
- Result: DC capacity exceeds verified AC capacity for **95.5%** of requests on case30, and only **3.9%** on case118, where it is conservative (median DC/AC 0.46). The error changes sign by network.



The two cases use different rating rules (see protocol), which confounds the comparison

Pooling is a cost change. The labels are identical.

- Requests share the network, the outage set and the operating state. Only the injection differs, so a sweep runs D × B stacked systems (requests × outages), request-major, with a per-request convergence mask.
- Rounds are set by the slowest request, not the sum. Memory is bounded by a fixed number of stacked systems.
- Bisection returns the largest scale actually verified secure, so a label can understate capacity but never overstate it.

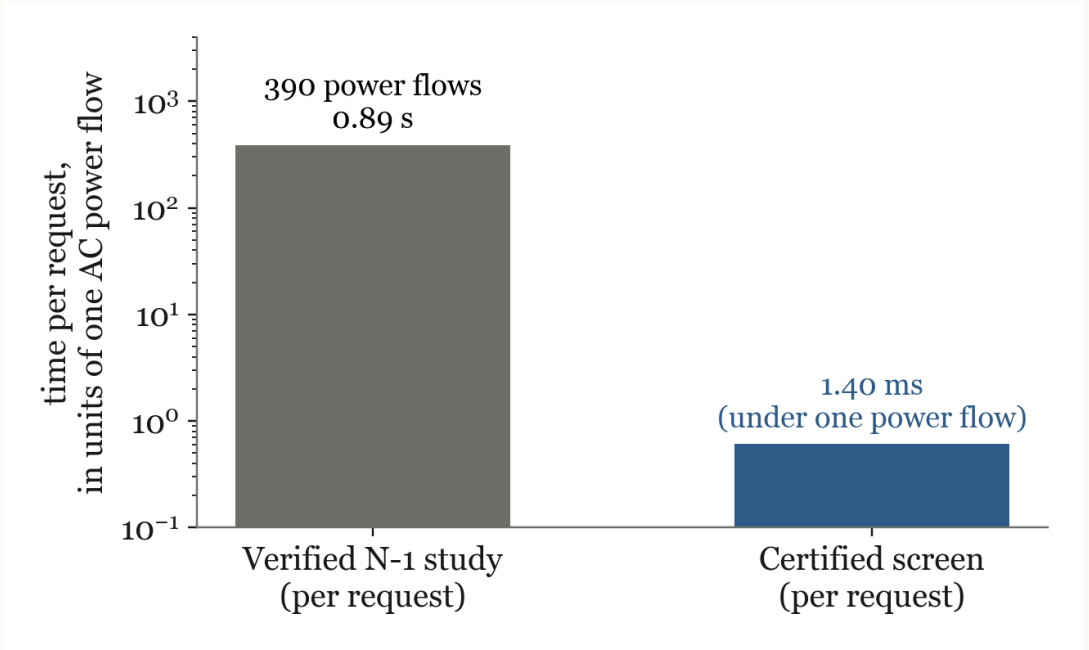
CHECK, CASE30	RESULT
Pooled vs one-at-a-time: capacity, binding mode, censoring, sweep evaluations	Equal (0% diff)
Queue lengths tested	4, 8, 16, 32
Speedup, 16-request check	2.18×
Speedup, over 80 labels	1.01×
Speedup scatter across queue lengths	0.68× to 1.77×

The pooling speedup is not established at these sizes. It was kept because labels are provably unchanged.

case30: 230 states, 38 studied outages, relative bisection tolerance 5e-03

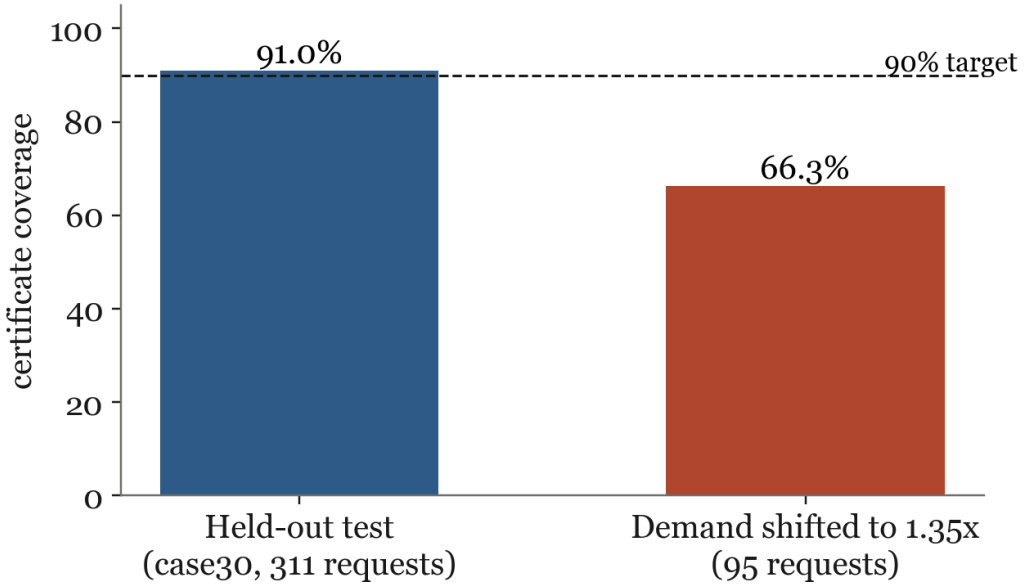
```
target  z = log(α_AC + ε₀) − log(α* + ε₀)
estimate α̂ = α* · e^(−ẑ)
```

- **Inputs:** per-branch sensitivities $b^{(c)}$ of the request, base post-contingency loadings, and the ratio of the binding constraint to the next tightest.
- **Architecture:** DeepONet. Branch arm is graph convolutions over the network with sensitivities as features. 183,428 parameters.
- At initialization it reproduces the DC closed form, and training only learns the correction.
- **Cost:** 1.40 ms per request against 390 power flows for a verified label. That is 390× in power-flow equivalents and 636× in wall clock.



```
q = ⌈(1−ε)(n+1)⌉-th smallest |ẑ − z| on calibration
certified α_c = α* · e^(−(ẑ + q))
defer if α_c < requested scale
```

CASE30, E = 10%	VALUE
Coverage, held-out test (311 requests)	91.0% (SE 1.6%)
Coverage across levels 80% to 99%	81.7% to 99.7%
Violation rate, point estimate vs certificate	47.9% vs 9.0%
Certified / verified capacity, median (p10)	0.90 (0.84)
DC / verified capacity, median	1.20



Split conformal: distribution-free, valid under exchangeability with the calibration set

The guarantee is within-distribution. Under demand shift it breaks.

Demand shifted to 1.35× nominal (outside the training grid, 95 requests)

- Coverage falls to **66.3%** against a 90% target (33.7% violation, SE 4.8%).
- 60.0% of shifted states fail their own base case before any request is added.
- Under shift, the defer rule has to do the work. That needs a shift detector, which isn't built yet.

case118 (12 screened outages of 177, 102 test requests)

- Coverage 94.1%. Certificate violation 5.9%; point estimate alone 68.6%.
- Certified / verified median 0.61, against 0.46 for DC.
- Label cost 3.17 s and 130 power flows. Screen is 130× cheaper.
- 26.5% of labels censored (bracket top still secure).

Shifted coverage on case118 rests on 6 admissible requests and is not relied on

The same certify-or-reject idea on a published grid foundation model.

- GENCO (IBM GridFM) predicts AC power flow after line outages. I ran a check on its published predictions for the **2,000-bus Texas synthetic grid**, N-1 test set.
- Each prediction is either certified by a power-flow check started from the learned answer, or rejected.
- Starting Newton from the learned answer also cuts the work: **2 iterations instead of 4**, 66 ms vs 121 ms.

989 OUTAGE SCENARIOS	RESULT
Certified	609
Scenarios that truly violate a limit	288, all rejected
Wrong answers certified	0
Raw predictions that look like a violation	907 (vs 288 true)

Code: github.com/sehajr-singhs/genco-certify

How the test cases were built.

ITEM	SETTING
Operating states	Demand scale {0.90, 1.00, 1.10, 1.20} × renewable output {0.50, 0.75, 1.00}, per-bus log-normal jitter
Requests	Unit power factor injection on 1 to 5 buses, sites ranked by load, log-normal weights
Splits, case30	869 train · 318 calibration · 311 test · 95 shifted (demand 1.35×)
Thermal ratings	case30: own ratings (median 27.8 MW). case118: worst unperturbed N-1 flow / 0.55
Voltage bands, case30	normal [0.941, 1.020] · contingency [0.835, 1.033]
Inadmissible states	train 9.2% · test 2.5% · shifted 60.0% (fail base case before any request)
Contingency screen, case118	Top 12 outages by the linear layer's own ranking. Forgone capacity measured: 1.00 of all-outage capacity

What has not been shown.

- **Network scale and realism.** 30 and 118 buses, with ratings I had to define. Capacities are self-consistent, not planning-grade. The screen itself hasn't run on a PJM-size model. The certify-or-reject check has run at 2,000 buses (slide 10).
- **Limits modeled.** No generator reactive limits. Voltage never bound in these runs, so the screen has only been exercised on thermal limits. The binding-limit classifier's 100% accuracy is a one-class result.
- **Distribution shift.** Coverage drops to 66.3% under demand shift. A separate learned tie-line screen (IEEE 39-bus) behaves the same way: 0.55 ms vs 256 ms per scenario and 91% congestion accuracy in distribution, but 28 to 32% loading error under N-1 topology changes.
- **End-to-end clearing time** is a projection. The defer rule is a rate, not a schedule.
- **No real queue data.** This is the gap a pilot would close.

Where a certified screen could fit in the study process.

ORDERING

Screen a full cycle first

Certified capacity for every request in milliseconds, so the requests most likely to bind are studied first.

DIAGNOSIS

Binding constraint up front

The linear layer names the binding outage and branch per request before any AC study runs.

RE-STUDIES

Re-screen on withdrawals

When requests drop out, the remaining queue can be re-screened immediately to see which earlier results may change.

In all three, the AC study remains the decision. Certified answers sit below verified capacity (median 0.90), and uncertified requests are deferred.

One historical cycle, offline, scored against PJM's own study results.

- **Data path:** whichever works under PJM's rules. Options: PJM runs the code internally and shares only aggregate metrics, a planning-grade public stand-in PJM considers representative, or PJM's process for outside researchers.
- **Metrics:** agreement with PJM's study outcomes, coverage at the chosen level, defer rate, studies that could have been skipped or reordered, and every case where the screen would have been wrong.

WEEKS	WORK
1 to 2	Model import (PSS/E RAW), match base-case power flow
3 to 4	AC labels on the cycle's requests, train operator, calibrate
5 to 6	Evaluation against study outcomes, written report

Questions

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